GCSE CHEMISTRY

Higher Tier – Paper 1 style

Topic 4.1: Atomic structure and the periodic table

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 180 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **171**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 171

0 1 This question is about the structure of atoms.

0 1 . 1 Complete the table to show the relative charge and the relative mass of each sub-atomic particle. **[3 marks]**

Particle	Relative charge	Relative mass
Proton		
Neutron		
Electron		

0 1 . 2 Explain why an atom has no overall charge. **[2 marks]**

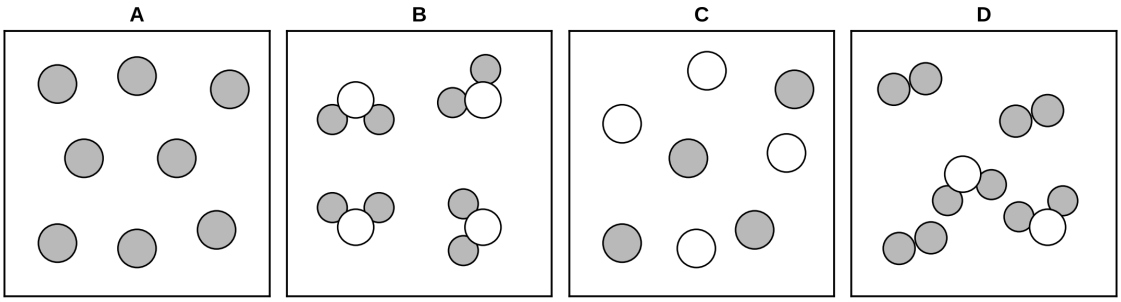
0 1 . 3 The radius of an atom is about 1×10^{-10} m. The radius of its nucleus is about 1×10^{-14} m. **[1 mark]**
How many times larger is the radius of the atom than the radius of the nucleus?

0 1 . 4 Where is most of the mass of an atom found? **[1 mark]**

*Tick **one** box.*

- ☐ In the electrons
- ☐ In the nucleus
- ☐ In the outer shell
- ☐ Spread evenly through the atom

0 2 The diagrams show the particles in four different substances, **A**, **B**, **C** and **D**.



Key: grey circle = atom of one element white circle = atom of a different element

0 2 . 1 Which diagram shows an element? Write the letter. **[1 mark]**

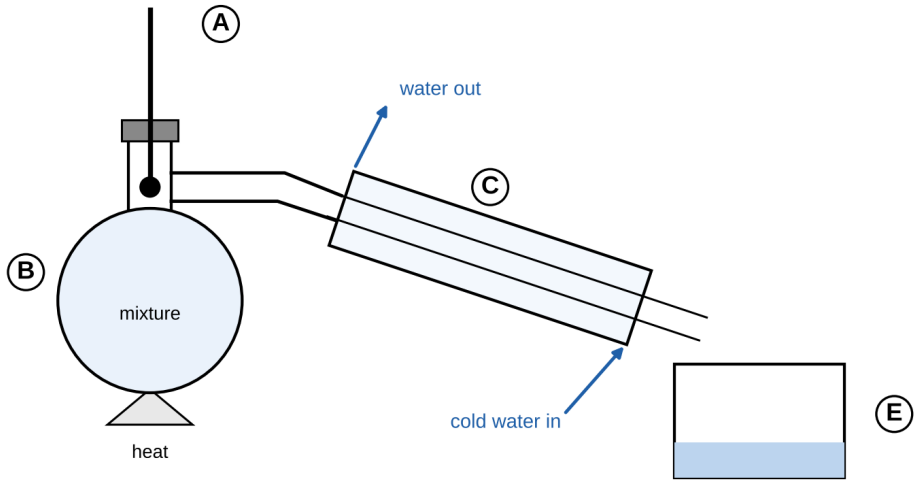
0 2 . 2 Which diagram shows a compound? Write the letter. **[1 mark]**

0 2 . 3 Which diagram shows a mixture of two elements? Write the letter. **[1 mark]**

0 2 . 4 Diagram **D** shows a mixture of an element and a compound. **[2 marks]**
Explain how you can tell that **D** is a mixture.

0 2 . 5 Water is a compound of hydrogen and oxygen. A mixture of hydrogen gas and oxygen gas is not the same as water. **[2 marks]**
Give **two** differences between the mixture of hydrogen and oxygen and the compound water.

0 3 A student sets up the apparatus shown to obtain pure water from sea water (salt dissolved in water).



0 3 . 1 Name the pieces of apparatus labelled **A** and **C**. **[2 marks]**

0 3 . 2 The cold water enters the condenser at the **lower** end and leaves at the upper end. **[2 marks]**
Explain why.

0 3 . 3 The thermometer bulb is placed level with the side arm and not in the liquid. **[1 mark]**
Suggest why.

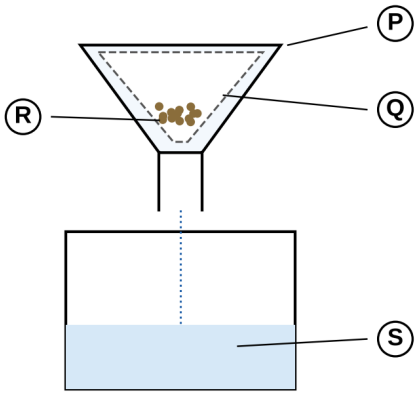
0 3 . 4 Explain how this apparatus produces pure water from sea water. **[3 marks]**

0 3 . 5 A mixture of ethanol (boiling point 78 °C) and water (boiling point 100 °C) is separated using **fractional** distillation. **[3 marks]**

Describe how the ethanol is separated from the water.

0 4 Rock salt is a mixture of salt (sodium chloride) and sand. Salt dissolves in water. Sand does not dissolve in water.

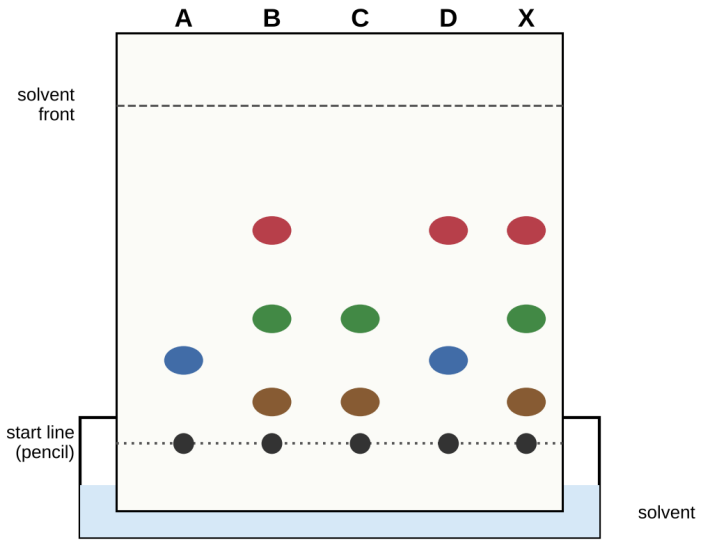
A student separates the salt from the sand. The diagram shows part of the apparatus.



0 4 . 1 Name the apparatus labelled **P**, and name the substance labelled **R** in this experiment. **[2 marks]**

0 4 . 2 *Describe how the student can obtain dry crystals of salt from the rock salt. Include the names of the processes used. **[6 marks]**

0 5 A forged cheque was found. A forensic scientist used paper chromatography to find out which of four pens, **A**, **B**, **C** and **D**, was used. She compared the ink from the cheque (**X**) with the ink from each pen. The chromatogram is shown.



0 5 . 1 The start line was drawn in pencil. Give a reason why pencil was used and not ink. **[1 mark]**

0 5 . 2 At the start of the experiment the level of the solvent must be **below** the start line. **[1 mark]**

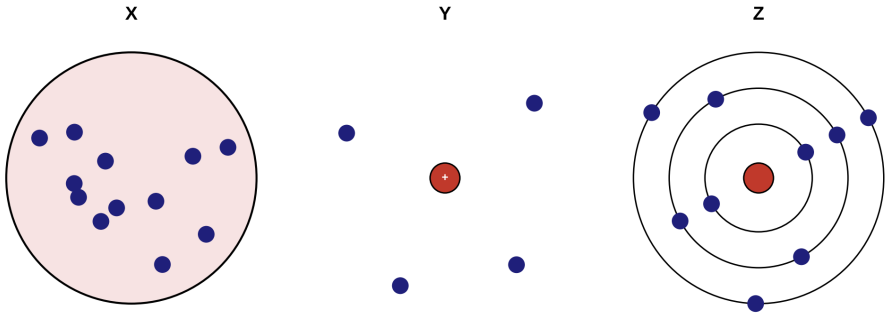
Explain why.

0 5 . 3 Which pen contains a pure substance? Give a reason for your answer. **[2 marks]**

0 5 . 4 Which pen was used to write the forged cheque? Explain how you know. **[2 marks]**

0 5 . 5 Explain why the different substances in an ink separate on the chromatography paper. **[2 marks]**

0 6 Scientists' ideas about the structure of the atom have changed over time.
Diagrams **X**, **Y** and **Z** show three different models of the atom.

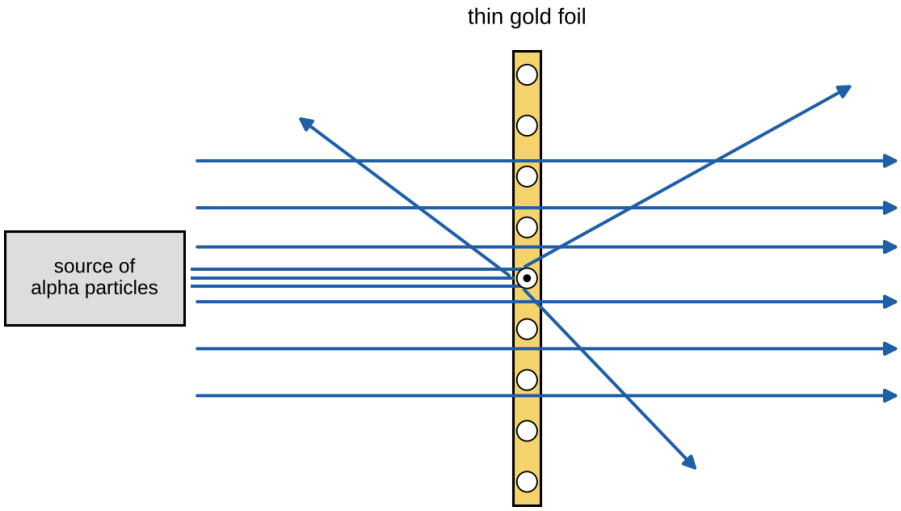


0 6 . 1 Which diagram shows the plum pudding model? Write the letter. **[1 mark]**

0 6 . 2 Describe the plum pudding model of the atom. **[2 marks]**

0 6 . 3 Diagram **Z** shows the model developed by Niels Bohr. Describe what this model shows about electrons. **[2 marks]**

0 7 In the alpha particle scattering experiment, positively charged alpha particles were fired at very thin gold foil. The diagram shows some of the paths taken by the alpha particles.



0 7 . 1 Describe what the scientists observed. **[3 marks]**

0 7 . 2 Explain what each observation shows about the structure of the atom. **[4 marks]**

0 7 . 3 The results of this experiment led scientists to replace the plum pudding model. **[1 mark]**
Suggest why scientists accepted the new model.

0 7 . 4 Later, James Chadwick provided evidence for the existence of a new particle in the nucleus. Name this particle. **[1 mark]**

0 8 This question is about atoms, ions and isotopes.

0 8 . 1 Complete the table. **[5 marks]**

Atom or ion	**Number of protons**	**Number of neutrons**	**Number of electrons**
^{35}Cl (atomic number 17)			
^{37}Cl (atomic number 17)			
$^{27}Al^{3+}$ (atomic number 13)			
$^{16}O^{2-}$ (atomic number 8)			
$^{39}K^{+}$ (atomic number 19)			

0 8 . 2 The atoms in the table of chlorine, ^{35}Cl and ^{37}Cl, are isotopes. **[2 marks]**
What is meant by the term **isotopes**?

0 8 . 3 Isotopes of the same element have the same chemical properties. Explain why. **[2 marks]**

0 9 Relative atomic mass (A_r) takes into account the abundance of the isotopes of an element.

0 9 . 1 A sample of chlorine contains 75% of chlorine-35 atoms and 25% of chlorine-37 atoms. **[3 marks]**
Calculate the relative atomic mass of chlorine in this sample.

Answer = _____

0 9 . 2 Magnesium has three isotopes. **[3 marks]**

magnesium-24: 79% magnesium-25: 10% magnesium-26: 11%

Calculate the relative atomic mass of magnesium. Give your answer to **1 decimal place**.

Answer = _____

0 9 . 3 Boron has two isotopes, boron-10 and boron-11. The relative atomic mass of boron is 10.8. **[3 marks]**
Calculate the percentage of boron-11 atoms in a sample of boron.

Answer = _____ %

0 9 . 4 Gallium has two isotopes, gallium-69 and gallium-71. The relative atomic mass of gallium is 69.8. **[3 marks]**
Calculate the percentage abundance of gallium-69.

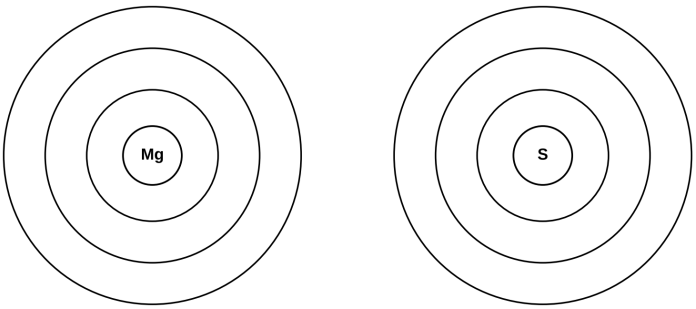
Answer = _____ %

1 0 This question is about electronic structure.

1 0 . 1 Write the electronic structure of each atom. **[3 marks]**

sodium (atomic number 11) _____
chlorine (atomic number 17) _____
calcium (atomic number 20) _____

1 0 . 2 Complete the diagrams to show the electronic structures of magnesium (atomic number 12) and sulfur (atomic number 16). Draw crosses (×) to show the electrons. **[2 marks]**



1 0 . 3 Sodium and potassium have similar chemical properties. **[2 marks]**
Explain why, using their electronic structures.

1 0 . 4 An element has the electronic structure 2,8,5. **[3 marks]**
Name the element and give its group and period in the periodic table.

1 1 This question is about the development of the periodic table.

1 1 . 1 Newlands arranged the known elements in order of atomic weight in 1864. **[1 mark]**
Many scientists did not accept his 'law of octaves'.
Give **one** reason why.

1 1 . 2 In 1869 Mendeleev published a periodic table. State **two** ways in which he arranged the elements. **[2 marks]**

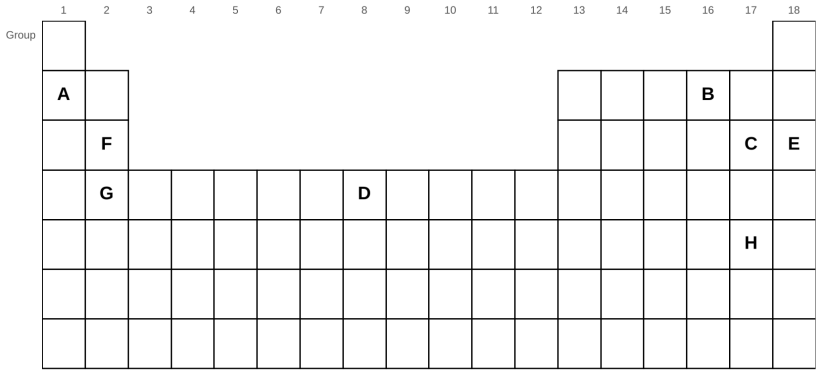
1 1 . 3 Mendeleev left gaps in his table. **[2 marks]**
Give **two** reasons why.

1 1 . 4 Mendeleev predicted that an element he called eka-silicon existed. Years later germanium was discovered and its properties matched his predictions. **[2 marks]**
Explain why this discovery was important.

1 1 . 5 Tellurium (A_r = 128) has a higher relative atomic mass than iodine (A_r = 127). In the modern periodic table tellurium is placed **before** iodine. Explain why. **[2 marks]**

1 1 . 6 *Mendeleev's table was accepted but Newlands' was not. Evaluate the reasons why. **[4 marks]**

1 2 The diagram shows the outline of the periodic table. The letters **A** to **H** are **not** the chemical symbols of the elements.



1 2 . 1 Which letter is an alkali metal? **[1 mark]**

1 2 . 2 Which letter is a noble gas? **[1 mark]**

1 2 . 3 Which **two** letters are halogens? **[1 mark]**

1 2 . 4 Which letter is a transition metal? **[1 mark]**

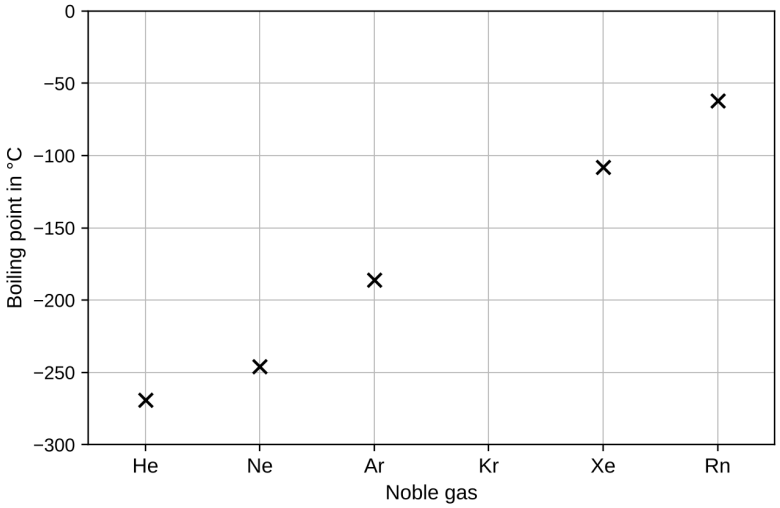
1 2 . 5 Which **two** letters are elements with similar chemical properties because they have two electrons in their outer shell? **[1 mark]**

1 2 . 6 Which letter is a non-metal that forms an acidic oxide? (Choose from **B** or **E**.) **[1 mark]**

1 2 . 7 Atoms of metals react to form positive ions. **[2 marks]**
Explain how a metal atom forms a positive ion.

1 2 . 8 Give **two** physical properties of non-metals. **[2 marks]**

1 3 This question is about the noble gases in Group 0. The graph shows the boiling points of five noble gases. The boiling point of krypton is missing.



1 3 . 1 Draw a line of best fit through the points. **[1 mark]**

1 3 . 2 Use your line to estimate the boiling point of krypton. **[1 mark]**

1 3 . 3 Describe the trend in boiling points down Group 0. **[1 mark]**

1 3 . 4 The noble gases are unreactive. **[2 marks]**
Explain why.

1 3 . 5 Helium is used in party balloons. Hydrogen is less dense than helium, but hydrogen is not used. **[2 marks]**
Explain why helium is used instead of hydrogen.

1 4 This question is about the alkali metals in Group 1.

1 4 . 1 A small piece of potassium is added to water. Give **two** observations. **[2 marks]**

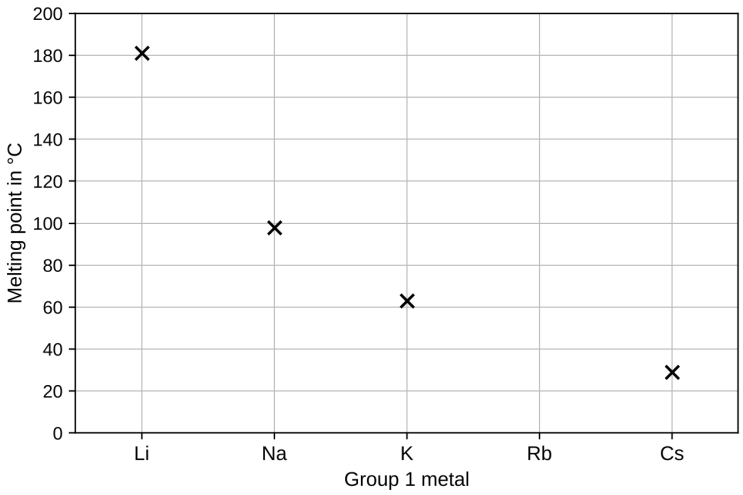
1 4 . 2 Sodium reacts with water. Write the **word equation** and the **balanced symbol equation** for the reaction. **[3 marks]**

1 4 . 3 Universal indicator is added to the solution formed when lithium reacts with water. **[2 marks]**
What colour does the indicator turn and why?

1 4 . 4 Potassium is more reactive than lithium. Explain this trend in reactivity **down** Group 1. **[4 marks]**

1 4 . 5 The graph shows the melting points of some Group 1 metals. Rubidium is missing. **[2 marks]**

Predict the melting point of rubidium.



1 4 . 6 Group 1 metals are stored in oil. Give a reason why. **[1 mark]**

1 4 . 7 Sodium reacts with chlorine to form sodium chloride. Write a balanced symbol equation for this reaction. **[2 marks]**

1 5 This question is about the halogens in Group 7.

1 5 . 1 Complete the table to show the colour and the state of chlorine, bromine and iodine at room temperature. **[3 marks]**

Halogen	**Colour**	**State at room temperature**
Fluorine	pale yellow	gas
Chlorine		
Bromine		
Iodine		

1 5 . 2 Astatine is below iodine in Group 7. **[2 marks]**
Predict the state of astatine at room temperature and give a reason for your answer.

1 5 . 3 Halogens exist as molecules made of two atoms, for example Cl_2. **[1 mark]**
What type of bond joins the atoms in a chlorine molecule?

1 5 . 4 A student adds each halogen to solutions of potassium halides. The table shows the results. **[3 marks]**
Put chlorine, bromine and iodine in order of reactivity (most reactive first) and explain how the results show this order.

Halogen added	**potassium chloride solution**	**potassium bromide solution**	**potassium iodide solution**
Chlorine	no change	orange solution	brown solution
Bromine	no change	no change	brown solution
Iodine	no change	no change	no change

1 5 . 5 Chlorine reacts with potassium bromide solution. Write a word equation and a balanced symbol equation for the reaction. **[3 marks]**

1 5 . 6 Fluorine is more reactive than chlorine. **[3 marks]**
Explain why, in terms of electronic structure.

1 5 . 7 Iron wool burns in chlorine. Write a word equation and a balanced symbol equation for the reaction. Iron(III) chloride is formed. **[3 marks]**

1 6 The table shows some information about sodium and two transition metals.

Metal	Melting point in °C	Density in g/cm^3	Reaction with water
Sodium	98	0.97	fizzes and melts
Iron	1538	7.87	very slow (rusts)
Copper	1085	8.96	no reaction

1 6 . 1 Give **three** ways that the transition metals are different from sodium (a Group 1 metal). **[3 marks]**

1 6 . 2 Transition metal compounds are often coloured. Give **one** example of a coloured transition metal compound or ion. **[1 mark]**

16.3 Iron is used as a catalyst in the Haber process. What is a catalyst? **[2 marks]**

16.4 Iron forms Fe^{2+} and Fe^{3+} ions. What does this show about transition metals? **[1 mark]**

16.5 Copper is used in electrical wiring. Give **two** properties of copper that make it suitable. **[2 marks]**

17 Stretch questions (grade 7–9).

17.1 *Compare the physical properties and the chemical reactions of the Group 1 metals with those of the transition metals. **[6 marks]**

1 7 . 2 *Describe and explain the trend in reactivity of the Group 7 elements. Use ideas about electronic structure and use displacement reactions as evidence. **[6 marks]**

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.1: Atomic structure and the periodic table

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	proton: +1 and 1 ; neutron: 0 and 1 ; electron: −1 and very small / 1/1840 / negligible / 0.0005 ;	One mark for each particle (both values needed). Allow '1/2000' for the mass of an electron. Do not accept '0' as the mass of an electron.	3	AO1 4.1.1.4
01.2	the number of protons equals the number of electrons ; positive and negative charges cancel / equal and opposite charges ;	Do not accept 'neutrons cancel out the charge'.	2	AO1 4.1.1.4
01.3	10 000 / 1×10^4 / 10^4	Allow 'ten thousand'.	1	AO2 4.1.1.5
01.4	nucleus	Tick the box next to 'In the nucleus'.	1	AO1 4.1.1.5
02.1	A		1	AO1 4.1.1.1
02.2	B		1	AO1 4.1.1.1
02.3	C		1	AO1 4.1.1.2
02.4	(it contains) different substances / more than one type of particle / both molecules of an element and molecules of a compound ; not chemically joined (to each other) / not in a fixed ratio / can be separated by physical methods ;	Do not accept 'it contains two elements' on its own - a compound also contains two elements.	2	AO2 4.1.1.2
02.5	in the compound the atoms are chemically bonded / in the mixture they are not ; the mixture can be separated by physical methods / the compound can only be separated by chemical reactions ; the compound has a fixed composition / the mixture can have any proportions ; the compound has different properties to the elements / the mixture keeps the properties of its parts (any two) ;	Any two of the points. Do not accept differences that only describe the state (liquid / gas).	2	AO1 4.1.1.2
03.1	A: thermometer ; C: (Liebig) condenser ;	Do not accept 'cooler' or 'tube' for C.	2	AO1 4.1.1.2
03.2	so the condenser is kept full of water / no air gaps ; water flows in the opposite direction to the vapour so cooling is more efficient / the coldest water meets the vapour that has cooled most ;	Allow 'countercurrent flow' for the second mark.	2	AO2 4.1.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	to measure the temperature of the vapour (leaving) / to measure the boiling point of the substance that is distilling	Do not accept 'to stop it breaking'.	1	AO3 4.1.1.2
03.4	the sea water is heated and the water evaporates / boils (at about 100 °C) ; the salt stays behind in the flask because it has a much higher boiling point ; the water vapour condenses in the condenser and is collected as liquid water ;	Do not accept 'the salt evaporates' or 'the water melts'.	3	AO1 4.1.1.2
03.5	the mixture is heated and the ethanol (lower boiling point) evaporates first ; the ethanol vapour rises up the fractionating column / water vapour condenses and runs back down ; ethanol vapour reaches the top at 78 °C, then condenses and is collected, leaving the water behind ;	Allow 'keep the temperature at 78 °C' for the third mark.	3	AO1 4.1.1.2
04.1	P: (filter) funnel ; R: sand ;	R is the residue.	2	AO1 4.1.1.2
04.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • add water to the mixture and stir so the salt dissolves • filter the mixture: sand is the residue (stays on the filter paper) • salt solution is the filtrate (passes through) • heat the filtrate (in an evaporating basin) to evaporate some of the water until crystals begin to form / saturated solution • leave to cool / leave to crystallise slowly • filter off the crystals and dry them (in a warm oven or between filter paper)	Level of response question: marks depend on the quality and order of the answer as well as the number of points.	6	AO1 4.1.1.2
05.1	pencil is insoluble (in the solvent) / pencil will not run / ink would dissolve and spread up the paper with the sample	Do not accept 'pencil is easier to rub out'.	1	AO1 4.1.1.2
05.2	so the samples do not dissolve into the solvent in the beaker / so the spots are not washed off the paper		1	AO2 4.1.1.2
05.3	A ; it produced only one spot / one (colour) substance ;	The mark for the reason needs the idea of a single spot.	2	AO3 4.1.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	B ; spots in the same positions / same number of spots at the same heights as X ;	Do not accept 'it has the same colours' without the idea of the same positions.	2	AO3 4.1.1.2
05.5	different substances have different solubilities in the solvent ; the more soluble a substance, the further it travels up the paper (with the solvent) ;	Allow 'different attraction to the paper and to the solvent' for the first mark. Do not accept 'different weights' on its own.	2	AO1 4.1.1.2
06.1	X		1	AO1 4.1.1.3
06.2	a ball / sphere of positive charge ; with (negative) electrons embedded / spread through it ;	Do not accept 'positive electrons'.	2	AO1 4.1.1.3
06.3	electrons orbit the nucleus ; in fixed shells / energy levels at set distances ;		2	AO1 4.1.1.3
07.1	most alpha particles passed straight through the foil ; some / a few were deflected (through small angles) ; very few / about 1 in 10 000 bounced back / were deflected through large angles ;		3	AO1 4.1.1.3
07.2	most pass straight through so the atom is mostly empty space ; some are deflected because the nucleus is positive and repels the positive alpha particles ; very few bounce back because the nucleus is very small / dense (contains most of the mass) ; so the plum pudding model was wrong / a small, positive central nucleus exists ;	A mark for each observation linked with its explanation.	4	AO2 4.1.1.3
07.3	the plum pudding model could not explain the results / the new model explained the results (of the experiment)		1	AO3 4.1.1.3
07.4	neutron		1	AO1 4.1.1.3
08.1	^{35}Cl: 17 protons, 18 neutrons, 17 electrons ; ^{37}Cl: 17 protons, 20 neutrons, 17 electrons ; $^{27}Al^{3+}$: 13 protons, 14 neutrons, 10 electrons ; $^{16}O^{2-}$: 8 protons, 8 neutrons, 10 electrons ; $^{39}K^{+}$: 19 protons, 20 neutrons, 18 electrons ;	One mark for each row (all three numbers correct). Common error: neutrons = mass number − atomic number. Electrons change only when the ion is charged.	5	AO2 4.1.1.4
08.2	atoms of the same element / with the same number of protons / same atomic number ; with different numbers of neutrons / different mass numbers ;	Do not accept 'different atomic mass' without the idea of different neutrons.	2	AO1 4.1.1.4
08.3	they have the same number of electrons / same electronic structure ; chemical reactions depend on (the number of) electrons (in the outer shell) ;	Do not accept 'same number of neutrons'.	2	AO2 4.1.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.1	(75 × 35) + (25 × 37) ; = 3550 ; 3550 ÷ 100 = 35.5 ;	Correct answer of 35.5 gains all 3 marks with or without working. Common error: dividing by 2 (the number of isotopes) instead of by 100.	3	AO2 4.1.1.6
09.2	(79 × 24) + (10 × 25) + (11 × 26) ; = 2432 (÷ 100 = 24.32) ; 24.3 ;	Allow 24.3 with no working for 3 marks. 24.32 scores 2 marks (not rounded to 1 d.p.).	3	AO2 4.1.1.6
09.3	let x = % of B-11: (10 × (100 − x) + 11x) ÷ 100 = 10.8 ; 1000 + x = 1080 ; x = 80 (%) ;	Allow 80 with no working for 3 marks. Allow an alternative method, e.g. 10.8 is 0.8 above 10, so 80% is boron-11.	3	AO2 4.1.1.6
09.4	let x = % of Ga-69: (69x + 71 × (100 − x)) ÷ 100 = 69.8 ; 7100 − 2x = 6980 ; x = 60 (%) ;	Allow 60 with no working for 3 marks. This is a grade 8–9 question: practise setting up the algebra.	3	AO2 4.1.1.6
10.1	sodium: 2,8,1 ; chlorine: 2,8,7 ; calcium: 2,8,8,2 ;	Allow 2.8.1 etc. Common error: calcium is 2,8,8,2 (the third shell holds 8 before the fourth shell starts).	3	AO1 4.1.1.7
10.2	magnesium: 2 electrons in first shell, 8 in second, 2 in third ; sulfur: 2 electrons in first shell, 8 in second, 6 in third ;	One mark for each atom correct.	2	AO1 4.1.1.7
10.3	both have one electron in their outer shell ; so both lose one electron (to form 1+ ions) in the same way ;	Allow 'same number of outer electrons'.	2	AO2 4.1.2.2
10.4	phosphorus ; Group 5 (1 mark: five outer electrons) ; Period 3 (1 mark: three shells) ;	Allow Group 15 (modern numbering). Link: group number = outer electrons; period number = number of occupied shells.	3	AO2 4.1.2.2
11.1	elements with different properties were placed in the same group / pattern broke down after calcium / he did not leave gaps for undiscovered elements / only worked for the first 17 elements	Any one.	1	AO1 4.1.2.2
11.2	in order of atomic weight (mass) ; elements with similar properties were put in the same column / group ;	Do not accept 'in order of atomic number' - that is the modern table.	2	AO1 4.1.2.2
11.3	he predicted that undiscovered elements existed ; so that elements with similar properties stayed in the same group ;		2	AO1 4.1.2.2
11.4	it showed his predictions were correct ; so scientists gained confidence in / accepted his table ;		2	AO3 4.1.2.2
11.5	modern table is in order of atomic number (not atomic mass) ; iodine has properties similar to the halogens (Group 7) / tellurium to Group 6 ;	Mendeleev also swapped these two elements, because the properties did not fit the order of atomic weight.	2	AO3 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.6	**Level 2 (3–4 marks)**: A clear answer covering the key ideas with correct reasoning linking them. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Mendeleev left gaps for undiscovered elements and predicted their properties • the predictions were tested and found to be correct (germanium, gallium), giving evidence • Mendeleev swapped some elements where the properties did not fit atomic weight • Newlands put dissimilar elements together and did not leave gaps • Newlands' pattern failed after the first 17 elements or so		4	AO3 4.1.2.2
12.1	A		1	AO1 4.1.2.1
12.2	E		1	AO1 4.1.2.1
12.3	C and H (both needed)		1	AO1 4.1.2.1
12.4	D		1	AO1 4.1.3
12.5	F and G (both needed)		1	AO2 4.1.2.2
12.6	B	B (Group 6) is a non-metal whose oxide dissolves to form an acidic solution; E is a noble gas which does not form an oxide.	1	AO2 4.1.2.3
12.7	the atom loses electron(s) (from its outer shell) ; to get a full outer shell / stable electronic structure ;	Do not accept 'gains protons'.	2	AO1 4.1.2.3
12.8	low melting and boiling points ; poor conductors of heat / electricity ; brittle (when solid) ; dull (when solid) ; low density (any two) ;	Any two. Do not accept 'not shiny' alone.	2	AO1 4.1.2.3
13.1	a smooth curve / straight line through (or very close to) all five points		1	AO2 4.1.2.4
13.2	between −140 °C and −160 °C	Allow answers read correctly from the student's own line within this range.	1	AO2 4.1.2.4
13.3	boiling point increases (down the group)		1	AO1 4.1.2.4
13.4	they have full outer shells of electrons ; so they do not need to lose, gain or share electrons / they have stable electronic structures ;	Allow 'helium has a full first shell of 2'.	2	AO1 4.1.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.5	helium is unreactive / non-flammable / will not burn or explode ; hydrogen is flammable / explosive ;	Allow 'helium is safe' for 1 mark only.	2	AO2 4.1.2.4
14.1	fizzing / bubbles / gas given off ; moves around on the surface ; melts into a ball ; lilac flame ; disappears / dissolves (any two) ;	Do not accept 'explodes' or 'turns purple' unless an indicator is mentioned.	2	AO1 4.1.2.5
14.2	sodium + water → sodium hydroxide + hydrogen ; $2Na + 2H_2O \rightarrow 2NaOH + H_2$; balanced ;	Mark 2 is for correct formulae; mark 3 is for the equation being balanced. Allow multiples (e.g. $4Na + 4H_2O \rightarrow 4NaOH + 2H_2$).	3	AO1 4.1.2.5
14.3	purple / dark blue ; the solution is alkaline / contains hydroxide ions (OH^-) ;	Allow 'blue-purple'. Do not accept 'red / orange'.	2	AO1 4.1.2.5
14.4	reactivity increases down the group ; the outer electron is further from the nucleus / there are more shells ; more shielding / the attraction between the nucleus and the outer electron is weaker ; so the outer electron is lost more easily ;	Common error: saying 'bigger atom, so more reactive' without the idea of weaker attraction on the outer electron. Do not accept 'more protons so more reactive'.	4	AO1 4.1.2.5
14.5	draws a curve through the points / uses the trend (decreasing down the group) ; value between 35 °C and 55 °C ;	Allow any value between 29 and 63 °C for 1 mark only.	2	AO2 4.1.2.5
14.6	to stop them reacting with oxygen / water (vapour) in the air	Do not accept 'because they are dangerous'.	1	AO1 4.1.2.5
14.7	$2Na + Cl_2 \rightarrow 2NaCl$; correct formulae and balanced ;	One mark for correct formulae, one mark for balancing.	2	AO1 4.1.2.5
15.1	chlorine: (pale) yellow-green, gas ; bromine: red-brown / orange-brown, liquid ; iodine: grey / dark grey / black, solid ;	One mark for each halogen (colour and state both correct). Allow 'purple vapour' in addition for iodine.	3	AO1 4.1.2.6
15.2	solid ; melting / boiling points increase down the group ;		2	AO2 4.1.2.6
15.3	covalent		1	AO1 4.1.2.6
15.4	chlorine, bromine, iodine ; chlorine displaced both bromide and iodide ions (from solution) ; bromine displaced iodide but not chloride, and iodine displaced neither ;	Principle: a more reactive halogen displaces a less reactive halogen from its salt.	3	AO3 4.1.2.6
15.5	chlorine + potassium bromide → potassium chloride + bromine ; $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$; balanced ;	Mark 2 for correct formulae, mark 3 for balancing.	3	AO1 4.1.2.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
15.6	a fluorine atom has fewer shells / the outer shell is closer to the nucleus ; less shielding / stronger attraction to the nucleus ; so an electron is gained more easily ;	Remember: reactivity **decreases** down Group 7 (opposite to Group 1).	3	AO2 4.1.2.6
15.7	iron + chlorine → iron(III) chloride ; $2Fe + 3Cl_2 \rightarrow 2FeCl_3$; balanced ;	Allow multiples.	3	AO1 4.1.2.6
16.1	higher melting points ; higher density ; less reactive with water ; harder / stronger ; (any three) ;	Use data from the table: e.g. 'iron melts at 1538 °C but sodium melts at 98 °C'. Do not accept 'they are metals'.	3	AO3 4.1.3.1
16.2	copper(II) sulfate (blue) / iron(II) (green) / iron(III) (orange-brown) / potassium manganate(VII) (purple) etc.	Any suitable example with its colour.	1	AO1 4.1.3.2
16.3	a substance that speeds up a reaction ; and is not used up (itself) / unchanged at the end ;		2	AO1 4.1.3.2
16.4	they can form ions with different charges		1	AO2 4.1.3.2
16.5	good electrical conductor ; can be bent / drawn into wires (ductile / malleable) ; unreactive / does not corrode easily ;	Any two.	2	AO2 4.1.3.2
17.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Group 1: soft (cut with a knife), low density (Li, Na, K float on water), low melting points • transition metals: hard / strong, high density, high melting points (except mercury) • Group 1: very reactive, react vigorously with water to form alkaline hydroxides and hydrogen • transition metals: less reactive, react slowly or not at all with water and oxygen • Group 1 compounds are white / colourless in solution; transition metal compounds are often coloured • Group 1 forms only 1+ ions; transition metals form ions with different charges and can act as catalysts		6	AO1 4.1.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
17.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • reactivity decreases down the group (F > Cl > Br > I) • atoms gain one electron to form 1− ions / get a full outer shell • going down, the outer shell is further from the nucleus and there is more shielding • so the attraction for an incoming electron is weaker and an electron is gained less easily • evidence: chlorine displaces bromine from potassium bromide solution (orange / brown solution forms) • bromine does not displace chlorine from potassium chloride solution; a more reactive halogen displaces a less reactive one		6	AO1 4.1.2.6

Where marks are lost in Topic 4.1

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Relative atomic mass:** divide by 100 (total % abundance) and not by the number of isotopes. Write the working: (% × mass) + (% × mass) ÷ 100.
- ☐ **Protons, neutrons, electrons:** neutrons = mass number − atomic number. Ions change the *electron* count only (Al^{3+} has lost 3).
- ☐ **Plum pudding model:** the electrons are negative and embedded in a ball of positive charge. Never write 'positive electrons'.
- ☐ **Alpha scattering:** every observation needs a conclusion: pass straight through = mostly empty space; deflected = positive nucleus; bounce back = tiny dense nucleus.
- ☐ **Electronic structure:** 2,8,8,2 for calcium. The group number is the number of outer electrons; the period number is the number of shells.
- ☐ **Group 1 trend explanations:** always give three links: outer electron further from nucleus → more shielding / weaker attraction → electron lost more easily.
- ☐ **Group 7 is the opposite:** reactivity **decreases** down the group because it is harder to *gain* an electron. Do not mix up Group 1 and Group 7 trends.
- ☐ **Displacement:** name the more reactive halogen and say it displaces the less reactive one. Quote the observation (e.g. orange solution of bromine).
- ☐ **Chromatography:** pencil start line (insoluble), solvent below the line, same height = same substance, one spot = pure.
- ☐ **Distillation:** evaporates then *condenses*. Name the apparatus correctly (condenser, not 'cooler'). Water enters at the bottom of the condenser.
- ☐ **6-mark questions:** plan 6 separate points before writing, use the key words (dissolve, filter, residue, filtrate, evaporate, crystallise) and never stop part-way.